# Fundamentals of Deep Learning and Programming

## Lecture 1: Useful Python Libraries

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# Python

## What is __call__( ) ?

- In Python, functions are callable (you can do f(x)).
- Objects (instances of classes) are not callable by default.
- But if a class defines the special method __call__, then its instances become callable, like functions.

```
class MyAdder:
    def __init__(self, n):
        self.n = n

    def __call__(self, x):
        return x + self.n

adder = MyAdder(10)  # adder is an instance of MyAdder class
print(adder(5))      # 15   (calls adder.__call__(5))
```

```
15
```

- Because we defined __call__, we can use adder(5) as if adder were a function.
- Internally, Python executes adder.__call__(5).

# Python

## Why is __call__( ) useful ?

- It lets you write objects that behave like functions, but also carry internal state.
- This is exactly how PyTorch's nn.Module works.

- Inside nn.Module.__call__, PyTorch does extra work:
  - Calls .forward()
  - Applies post-processing

So you can write criterion(y_hat, y) instead of criterion.forward(y_hat, y)
– much cleaner!

Let `criterion` be an instance of `nn.MSELoss`:

$$\texttt{criterion} = \texttt{nn.MSELoss()}.$$

## Callable Object via `__call__`

In Python, when we write

$$\texttt{loss} = \texttt{criterion}(y_{\text{hat}}, y),$$

this does not call the constructor again. Instead, it invokes the special method

$$\texttt{criterion.\_\_call\_\_(y\_hat, y)}.$$

## Delegation to `forward`

For PyTorch modules, the method `__call__` internally delegates to the `forward` function:

$$\texttt{criterion.\_\_call\_\_(y\_hat, y)} \longrightarrow \texttt{criterion.forward}(y\_\text{hat}, y).$$

## Mean Squared Error Computation

Specifically, for `nn.MSELoss`, the forward method computes the mean squared error:

$$\mathcal{L}(y_{\text{hat}}, y) = \frac{1}{N} \sum_{i=1}^{N} (y_{\text{hat},i} - y_i)^2 .$$

## Summary

Thus,

$$\texttt{loss} = \texttt{criterion}(y_{\text{hat}}, y)$$

is equivalent to evaluating the MSE loss between the prediction $y_{\text{hat}}$ and the target $y$.

# Numpy

= **Num**erical **Python**

**conda install numpy**

**pip install numpy**

- **Multi-dimensional arrays (ndarray)** : Unlike Python lists, NumPy arrays store elements of the same type, allowing faster computation and less memory usage.
- **Mathematical operations**: A wide range of high-performance functions for linear algebra, statistics, and random number generation.
- **Vectorization**: Applying operations to **entire arrays at once**, making code more concise and much **faster**.
- **Easy Integration with other Python libraries** such as pandas, SciPy, scikit-learn, TensorFlow, and PyTorch.

# Numpy

## Create Arrays

performing **operations on entire arrays (vectors, matrices, tensors) at once**, instead of using explicit Python for loops.

```python
import numpy as np

# Create arrays
a = np.array([1, 2, 3])
b = np.array([[1, 2, 3],
              [4, 5, 6]])

print(a)               # [1 2 3]
print(b)               # 2 rows, 3 cols

print(a.ndim)          # number of axes: 1
print(b.ndim)          # number of axes: 2
print(b.shape)         # (2, 3) -> 2 rows, 3 columns
print(b.dtype)         # data type (e.g., int64)

# Make arrays of zeros/ones or a range
z = np.zeros((2, 3))           # shape 2x3, all zeros (float)
o = np.ones((3, 2), dtype=int) # shape 3x2, all ones (int)
r = np.arange(0, 10, 2)        # [0  2  4  6  8]
lin = np.linspace(0, 1, 5)     # [0.  0.25  0.5  0.75  1. ]
print(z, o, r, lin, sep="\n")
```

```
[1 2 3]
[[1 2 3]
 [4 5 6]]
1
2
(2, 3)
int64
```

```
[[0. 0. 0.]
 [0. 0. 0.]]
[[1 1]
 [1 1]
 [1 1]]
[0 2 4 6 8]
[0.   0.25 0.5  0.75 1.  ]
```

c02_numpy01.py

# Numpy

## Indexing & Slicing

```
B = [[10, 11, 12],
     [20, 21, 22],
     [30, 31, 32]]

print(B[0][1])
```

list

```
11
```

```
import numpy as np

A = np.array([[10, 11, 12],
              [20, 21, 22],
              [30, 31, 32]])

print(A[0, 1])     # row 0, col 1 -> 11
print(A[1])        # entire row 1 -> [20 21 22]
print(A[:, 2])     # entire col 2 -> [12 22 32]
print(A[0:2, 1:3]) # rows 0..1, cols 1..2 -> [[11 12],[21 22]]

# Boolean indexing (filtering)
mask = (A % 20 == 0)        # True where divisible by 20
print(mask)                 # [[False False False],[ True False False],[ True False False]]
print(A[mask])              # [20]
```

numpy

```
11
[20 21 22]
[12 22 32]
[[11 12]
 [21 22]]
```

```
[[False False False]
 [ True False False]
 [False False False]]
[20]
```

c02_numpy02.py

# Numpy

## Vectorization

performing **operations on entire arrays (vectors, matrices, tensors) at once**, instead of using explicit Python for loops.

```
a = list(range(5))
b = list(range(5))

result = [a[i] + b[i] for i in range(len(a))]  # without vectorization
```

list, for loop

```
[0, 2, 4, 6, 8]
```

```
import numpy as np
a = np.arange(5)
b = np.arange(5)
result = a + b   # vectorization
```

numpy

```
[0 2 4 6 8]
```

c02_numpy03.py

```
numbers = list(range(5))
squared = [x**2 for x in numbers]
print(squared)
```

list, for loop

```
[0, 1, 4, 9, 16]
```

```
numbers = np.arange(5)
squared = numbers ** 2  # vectorization
print(squared)
```

numpy

```
[ 0  1  4  9 16]
```

# Numpy

## Broadcasting

(different shapes that "fit")

```python
import numpy as np

A = np.array([[1, 2, 3],
              [4, 5, 6]])   # shape (2,3)

d = 10
print(A + d)
# [[11 12 13]
#  [14 15 16]]

print(type(d))    # <class 'int'>
print(type(A+d))  # <class 'numpy.ndarray'>
```

```
[[11 12 13]
 [14 15 16]]
<class 'int'>
<class 'numpy.ndarray'>
```

```python
import numpy as np

A = np.array([[1, 2, 3],
              [4, 5, 6]])   # shape (2,3)
b = np.array([10, 20, 30])  # shape (3,)

print(A + b)
# [[11 22 33]
#  [14 25 36]]
```

```
[[11 22 33]
 [14 25 36]]
```

```python
import numpy as np

A = np.array([[1, 2, 3],
              [4, 5, 6]])   # shape (2,3)
c = np.array([[100],        # shape (2,1)
              [200]])
print(A+ c)
# [[101 102 103]
#  [204 205 206]]
```

```
[[101 102 103]
 [204 205 206]]
```

c02_numpy04.py

# Numpy

## Axis

axis=0 go down columns (across rows)
axis=1 go across rows (across columns)

IMPORTANT

```python
import numpy as np

B = np.array([[1, 2, 3],
              [4, 5, 6]])

print(B.sum())              # 21
print(B.mean(), B.min(), B.max())  # 3.5 1 6
print(np.median(B), np.std(B))     # 3.5, standard deviation

# axis ----------------------------------------
print(B.sum(axis=0))        # column sums -> [5 7 9]
print(B.sum(axis=1))        # row sums -> [ 6 15]

# column-wise mean (axis=0)
col_means = B.mean(axis=0)
print("axis=0 (columns) mean:", col_means)   # [2.5 3.5 4.5]

# row-wise mean (axis=1)
row_means = B.mean(axis=1)
print("axis=1 (rows) mean:", row_means)      # [2. 5.]
```

```
21
3.5 1 6
3.5 1.707825127659933
```

```
[5 7 9]
[ 6 15]
```

```
axis=0 (columns) mean: [2.5 3.5 4.5]
```

```
axis=1 (rows) mean: [2. 5.]
```

c02_numpy05.py

# Numpy

## reshape( ), ravel( )

```
import numpy as np

C = np.arange(12)          # [0..11]
print(C.reshape(3, 4))     # shape (3,4)
print(C.reshape(-1, 6))    # -1 lets NumPy infer -> (2,6)
print(C.ravel())           # flat view if possible
```

```
[[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]
```

```
[[ 0  1  2  3  4  5]
 [ 6  7  8  9 10 11]]
```

```
[ 0  1  2  3  4  5  6  7  8  9 10 11]
```

c02_numpy06.py

# Numpy

## np.array([1., 2.]) is a 1-D array whose shape is (2,).

`b = np.array([1., 2.])` is **neither a row nor a column vector.** It's a **1-D array with shape (2,).**
In NumPy, a vector's "direction" only becomes clear when you make it 2-D array.

```
import numpy as np

A = np.array([[1., 2.],
              [3., 4.]])
b = np.array([1., 2.])

row = b[None, :]          # shape: (1, 2)  ← row vector
col = b[:, None]          # shape: (2, 1)  ← column vector

# Equivalent forms
row2 = b.reshape(1, -1)  # (1, 2)
col2 = b.reshape(-1, 1)  # (2, 1)

# Matrx multiplication
print(b @ A)     # (2,) @ (2,2) → (2,)   (behaves like a row in this context)
print(A @ b)     # (2,2) @ (2,) → (2,)   (behaves like a column in this
context)

print(row @ A)   # (1,2) @ (2,2) → (1,2)
print(A @ col)   # (2,2) @ (2,1) → (2,1)
```

```
[ 7. 10.]
[ 5. 11.]
```

```
[[ 7. 10.]]
```

```
[[ 5.]
 [11.]]
```

`row = b[None, :]`

None (= np.newaxis) tells NumPy to 'add a new axis (dimension).'

c02_numpy06.py

# Numpy

## np.stack( )

np.stack( ) → adds a new axis.

axis=0 go down columns (across rows)
axis=1 go across rows (across columns)

```
import numpy as np

X = np.array([1, 2, 3])
Y = np.array([4, 5, 6])

print(np.stack([X, Y], axis=0))  # [[1 2 3], [4 5 6]]
print(np.stack([X, Y], axis=1))  # [[1 4], [2 5], [3 6]]

print(np.vstack([X, Y]))   # [[1 2 3], [4 5 6]]
print(np.hstack([X, Y]))   # [1 2 3 4 5 6]
```

```
[[1 2 3]
 [4 5 6]]
```

```
[[1 4]
 [2 5]
 [3 6]]
```

```
[[1 2 3]
 [4 5 6]]
```

```
[1 2 3 4 5 6]
```

```
import numpy as np

X = np.array([1, 2, 3])
Y = np.array([4, 5, 6])

print(X.shape, Y.shape)

a = np.stack([X, Y], axis=0)
b = np.stack([X, Y], axis=1)
print(a.shape)
print(b.shape)
```

```
(3,) (3,)
(2, 3)
(3, 2)
```

c02_numpy06.py

# Numpy

## Linear algebra

c02_numpy06.py

```
import numpy as np

A = np.array([[2., 1.],
              [1., 3.]])
b = np.array([1., 2.])

# Solve A x = b
x = np.linalg.solve(A, b)
print("solution:", x)
```

```
solution: [0.2 0.6]
```

```
# Matrix multiply (dot product)
M = np.array([[1, 2],
              [3, 4]])
N = np.array([[10, 20],
              [30, 40]])

print(M * N)        # caution: element-wise multiplication
print(M @ N)         # matrix multiplication
print(np.dot(M, N))  # matrix multiplication (M @ N)

print(np.linalg.eig(M))    # eigenvalues & eigenvectors
```

```
[[ 10  40]
 [ 90 160]]
```

```
[[ 70 100]
 [150 220]]
```

```
[[ 70 100]
 [150 220]]
```

```
EigResult(eigenvalues=array([-0.37228132,  5.37228132]), eigenvectors=array([[-0.82456484, -0.41597356],
       [ 0.56576746, -0.90937671]]))
```

# Pandas

**= Spreadsheet using Python**

**conda install pandas** **pip install pandas**

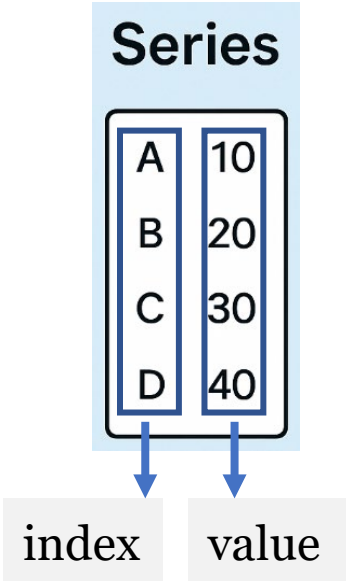
- **Structured data** : Works perfectly with tabular data (rows & columns). Similar to Excel, but much more powerful.
- **Fast computations** with the flexibility of Python**.**
- Widely used in **data science, machine learning,** and finance.

# Pandas

## Series

A **1D** labeled array

- A Series is one-dimensional, but **pandas adds labels (index) to each value**.
- This makes the data more powerful and easier to use.




```python
import numpy as np
import pandas as pd

# Create a Series from a NumPy array
data = np.array([10, 20, 30, 40])
s = pd.Series(data, index=["A", "B", "C", "D"])
print(s)

print(s.values) # If you want to see values
```

```
A    10
B    20
C    30
D    40
dtype: int64

[10 20 30 40]
```

c02_pandas01.py

When you print a Series, pandas shows a summary of the object:

- **Index (labels)** → on the left (A, B, C, D)
- **Values** → on the right (10, 20, 30, 40)
- dtype → at the bottom (int64)

# Pandas

## DataFrame

A **2D** labeled table

| Category | Age | weight |
|---|---|---|
| dog | 3 | 15 |
| cat | 2 | 10 |
| goat | 8 | 35 |

```
import pandas as pd

# Create a DataFrame from a dictionary
data = {
    "Category": ["dog", "cat", "goat"],
    "Age": [3, 2, 8],
    "weight": [15, 10, 35]
}

#converts this dictionary into a DataFrame (a table)
df = pd.DataFrame(data)
print(df)
```

```
  Category  Age  weight
0      dog    3      15
1      cat    2      10
2     goat    8      35
```

```
print(df["Age"])
```

```
0    3
1    2
2    8
Name: Age, dtype: int64
```

```
print(df["Age"].mean())  # Average of "Age"
```

```
4.333333333333333
```

c02_pandas01.py

- **Index** → labels for rows (here: 0, 1, 2).
- **Series** → each individual column of the DataFrame.
- **DataFrame** → a collection of multiple Series sharing the same index.

# Pandas

| Category | Age | weight |
|---|---|---|
| dog | 3 | 15 |
| cat | 2 | 10 |
| goat | 8 | 35 |

- **iloc** stands for integer-location based indexing

c02_pandas01.py

```
import pandas as pd

# Create a DataFrame from a dictionary
data = {
    "Category": ["dog", "cat", "goat"],
    "Age": [3, 2, 8],
    "weight": [15, 10, 35]
}
df = pd.DataFrame(data)

print(df.iloc[0])    # first row

print(df.iloc[1])    # second row

print(df.iloc[:,0])  # first column
```

```
Category    dog
Age           3
weight       15
Name: 0, dtype: object
```

```
Category    cat
Age           2
weight       10
Name: 1, dtype: object
```

```
0      dog
1      cat
2     goat
Name: Category, dtype: object
```

# Matplotlib

**= making charts and plots**

**conda install matplotlib**

**pip install matplotlib**

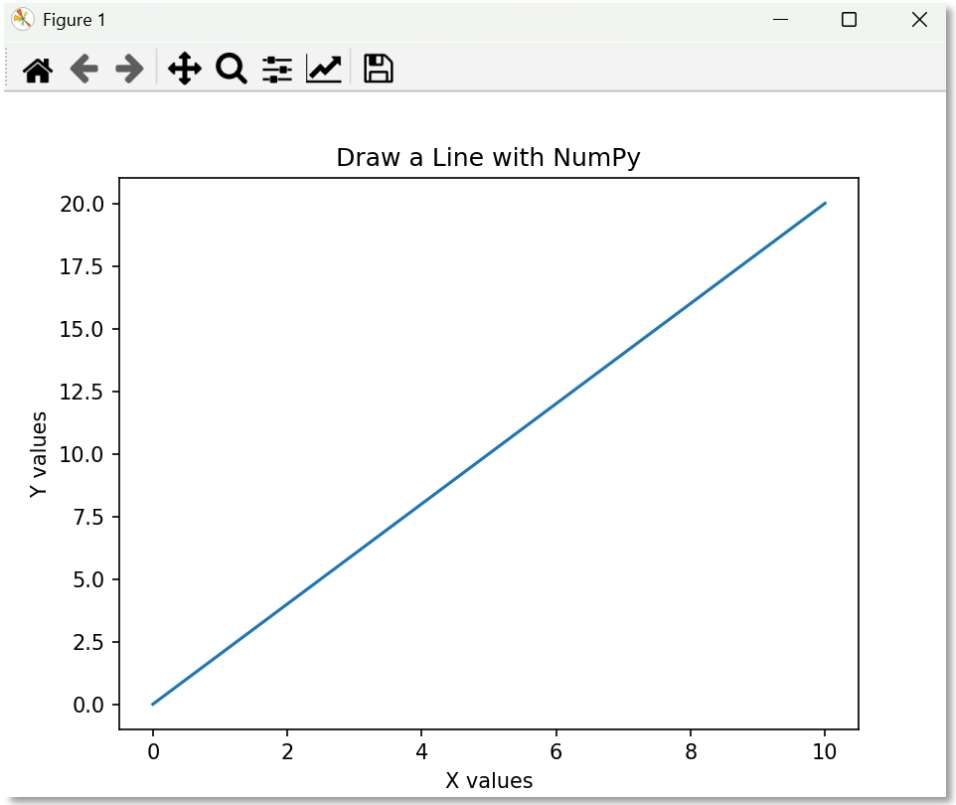
- You can make **line charts**, **bar charts**, **scatter plots**, **histograms**, and more.

# Matplotlib

## Draw a Line

```
import numpy as np
import matplotlib.pyplot as plt

# Create data with NumPy
x = np.linspace(0, 10, 100)   # 100 points from 0 to 10
y = 2 * x                     # y = 2x

plt.plot(x, y)

plt.title("Draw a Line with NumPy")
plt.xlabel("X values")
plt.ylabel("Y values")

plt.show()
```

c02_matplotlib01.py

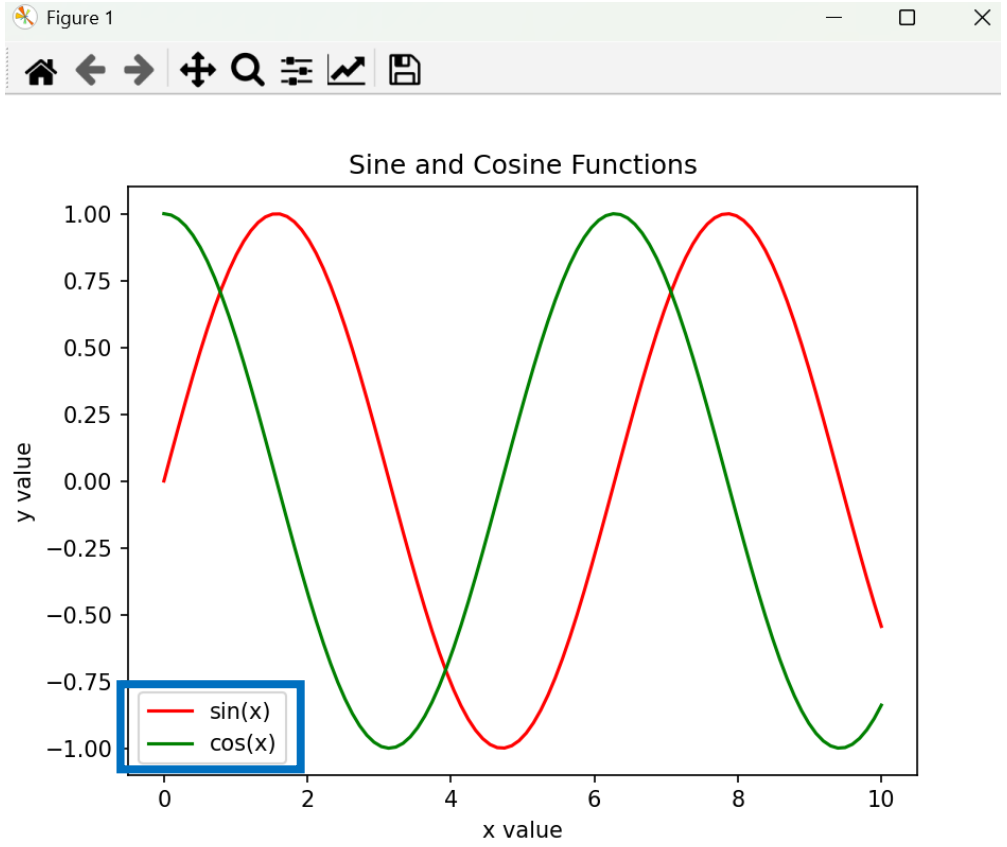


# Matplotlib

## Multiple Plots on the Same Graph

```
import numpy as np
import matplotlib.pyplot as plt

# Create data
x = np.linspace(0, 10, 100)
y1 = np.sin(x)
y2 = np.cos(x)

# Plot both
plt.plot(x, y1, label="sin(x)", color="red")
plt.plot(x, y2, label="cos(x)", color="green")

plt.xlabel("x value")
plt.ylabel("y value")
plt.title("Sine and Cosine Functions")
plt.legend()
plt.show()
```

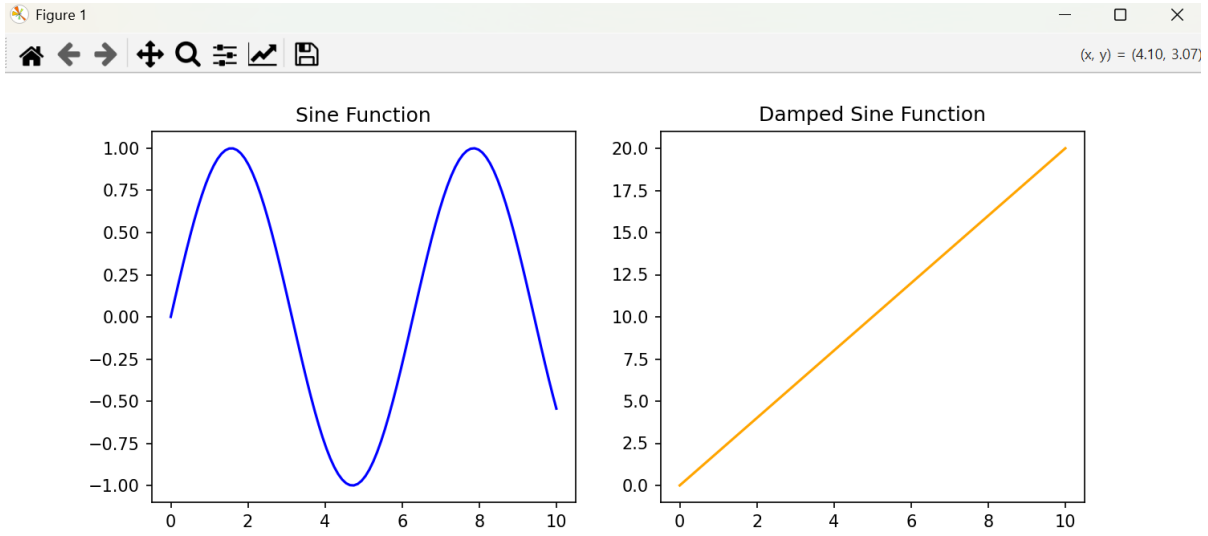
c02_matplotlib02.py



# Matplotlib

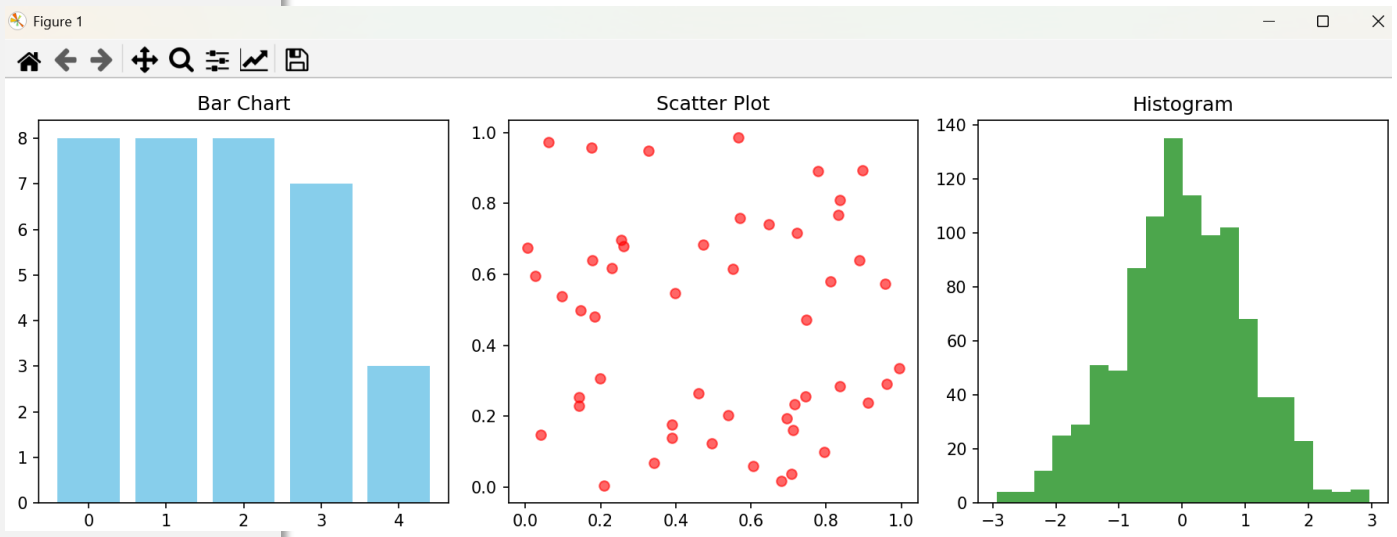
## Multiple Graphs Side by Side

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 10, 100)
y1 = np.sin(x)
y2 = 2 * x

# fig = whole figure (canvas),
# axes = subplots (small plots inside the figure)
# 1 row, 2 columns → two plots side by side
# figsize=(10, 4) → width=10 inches, height=4 inches
fig, axes = plt.subplots(1, 2, figsize=(10, 4))

# Left subplot
axes[0].plot(x, y1, color="blue")
axes[0].set_title("Sine Function")
axes[0].set_xlabel("x")
axes[0].set_ylabel("sin(x)")

# Right subplot
axes[1].plot(x, y2, color="red")
axes[1].set_title("Linear Function")
axes[1].set_xlabel("x")
axes[1].set_ylabel("2*x")

plt.tight_layout()  # auto-adjust spacing between
                    # subplots
plt.show()
```



# Matplotlib

```
import matplotlib.pyplot as plt
import numpy as np

# Sample data
x = np.arange(5)
y = np.random.randint(1, 10, size=5)
scatter_x = np.random.rand(50)
scatter_y = np.random.rand(50)
hist_data = np.random.randn(1000)

# Create figure with 3 subplots
fig, axes = plt.subplots(1, 3, figsize=(12, 4))

# Bar chart
axes[0].bar(x, y, color='skyblue')
axes[0].set_title("Bar Chart")

# Scatter plot
axes[1].scatter(scatter_x, scatter_y, c='red', alpha=0.6)
axes[1].set_title("Scatter Plot")

# Histogram
axes[2].hist(hist_data, bins=20, color='green', alpha=0.7)
axes[2].set_title("Histogram")

plt.tight_layout()
plt.show()
```




**creates a bar chart, scatter plot, and histogram**

c02_matplotlib03.py

# Basics

# Epoch, Batch, Iteration

| | |
|---|---|
| **One Epoch** | **One full training cycle** over the entire dataset |
| **Batch** | (=mini-batch) A subset of the dataset used in one update step |
| **Iteration** | Processing one batch |

Example

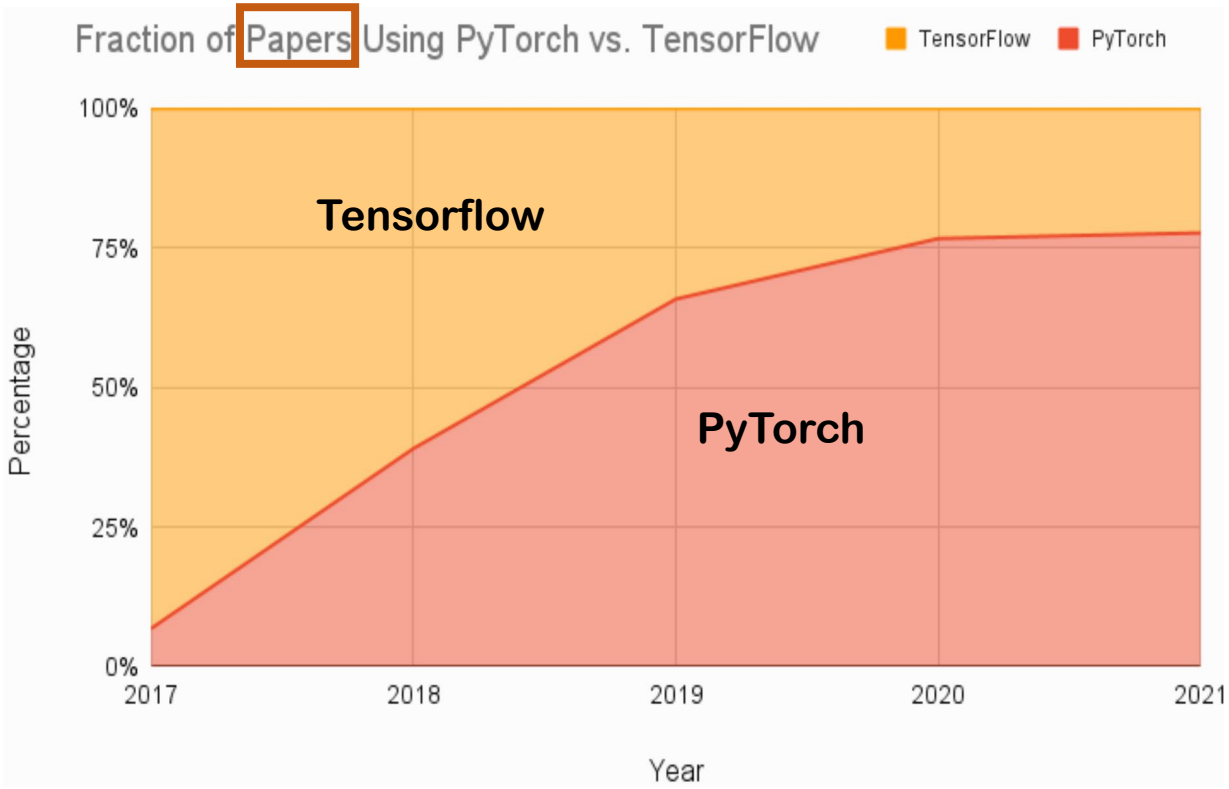
Suppose we have a dataset of **1,000 samples** and we choose **a batch size of 100**.

- **Batch size = 100** → Each batch contains 100 samples.
- Total dataset = 1,000 samples → That means there are 10 batches per epoch.
- So, 1 epoch = 10 iterations.

If we train the model for **5 epochs**:
The model will see the entire dataset 5 times.
Total iterations = **5 epochs** × 10 iterations/epoch = 50 iterations.

# PyTorch Basics

# PyTorch vs. TensorFlow



# Creating Tensor

A PyTorch tensor is a multi-dimensional array that is optimized for numerical computation.

## PyTorch

```
import torch

# Create tensors
a = torch.tensor([1, 2, 3])
b = torch.tensor([[1, 2, 3],
                  [4, 5, 6]])

print(a)                # tensor([1, 2, 3])
print(b)                # 2 rows, 3 cols

print(a.ndim)           # number of axes: 1
print(b.ndim)           # number of axes: 2
print(b.shape)          # torch.Size([2, 3]) -> 2 rows, 3 columns
print(b.dtype)          # data type (e.g., torch.int64)

# Make tensors of zeros/ones or a range
z = torch.zeros((2, 3))                  # shape 2x3, all zeros (float)
o = torch.ones((3, 2), dtype=torch.int32)   # shape 3x2, all ones (int)
r = torch.arange(0, 10, 2)                  # [0, 2, 4, 6, 8]
lin = torch.linspace(0, 1, steps=5)         # [0., 0.25, 0.5, 0.75, 1.]
print(z, o, r, lin, sep="\n")
```

```
tensor([1, 2, 3])
tensor([[1, 2, 3],
        [4, 5, 6]])
1
2
torch.Size([2, 3])
torch.int64
```

```
tensor([[0., 0., 0.],
        [0., 0., 0.]])
tensor([[1, 1],
        [1, 1],
        [1, 1]], dtype=torch.int32)
tensor([0, 2, 4, 6, 8])
tensor([0.0000, 0.2500, 0.5000, 0.7500, 1.0000])
```

c02_pytorch01.py

# PyTorch Tensor

1. Similar to NumPy arrays – they support **efficient numerical operations**.
2. **Automatic differentiation** – when used with requires_grad=True, tensors can **track computations for backpropagation in deep learning**.
3. **GPU acceleration** – tensors can be **easily moved to GPUs** ( .to("cuda") ) for faster computation.
4. Versatility – they can represent almost any type of data: text embeddings, image pixels, audio signals, etc.

# Indexing & Slicing

## PyTorch

```
import torch

A = torch.tensor([[10, 11, 12],
                  [20, 21, 22],
                  [30, 31, 32]])

print(A[0, 1].item()) # row 0, col 1 -> 11
print(A[1])           # entire row 1 -> tensor([20, 21, 22])
print(A[:, 2])        # entire col 2 -> tensor([12, 22, 32])
print(A[0:2, 1:3])    # rows 0..1, cols 1..2
                      #    -> tensor([[11, 12],[21, 22]])

# Boolean indexing (filtering)
mask = (A % 20 == 0)      # True where divisible by 20
print(mask)
print(A[mask])            # tensor([20])
```

```
11
tensor([20, 21, 22])
tensor([12, 22, 32])
tensor([[11, 12],
        [21, 22]])
```

c02_pytorch02.py

```
tensor([[False, False, False],
        [ True, False, False],
        [False, False, False]])
tensor([20])
```

## Numpy

```
import numpy as np

A = np.array([[10, 11, 12],
              [20, 21, 22],
              [30, 31, 32]])

print(A[0, 1])     # row 0, col 1 -> 11
print(A[1])        # entire row 1 -> [20 21 22]
print(A[:, 2])     # entire col 2 -> [12 22 32]
print(A[0:2, 1:3]) # rows 0..1, cols 1..2
                   #   -> [[11 12],[21 22]]

# Boolean indexing (filtering)
mask = (A % 20 == 0)  # True where divisible by 20
print(mask)
print(A[mask])        # [20]
```

```
11
[20 21 22]
[12 22 32]
[[11 12]
 [21 22]]
```

```
[[False False False]
 [ True False False]
 [False False False]]
[20]
```

# Vectorization

performing **operations on entire arrays (vectors, matrices, tensors) at once**, instead of using explicit Python for loops.

## PyTorch

```
import torch

a = torch.arange(5)    # tensor([0, 1, 2, 3, 4])
b = torch.arange(5)    # tensor([0, 1, 2, 3, 4])

result = a + b         # vectorization
print(result)                          tensor([0, 2, 4, 6, 8])

numbers = torch.arange(5)              tensor([ 0,  1,  4,  9, 16])
squared = numbers ** 2  # vectorization
print(squared)
```

c02_pytorch03.py

## Numpy

```
import numpy as np
a = np.arange(5)
b = np.arange(5)

result = a + b    # vectorization    [0 2 4 6 8]
print(result)

numbers = np.arange(5)
squared = numbers ** 2  # vectorization
print(squared)                                  [ 0  1  4  9 16]
```

## list, for loop

```
a = list(range(5))
b = list(range(5))

result = [a[i] + b[i] for i in range(len(a))]
# without vectorization          [0, 2, 4, 6, 8]
```

```
numbers = list(range(5))
squared = [x**2 for x in numbers]
# without vectorization

print(squared)          [0, 1, 4, 9, 16]
```

# Broadcasting (different shapes that "fit")

## PyTorch

```
import torch

A = torch.tensor([[1, 2, 3],
                  [4, 5, 6]])   # shape (2,3)

d = 10
print(A + d)
# tensor([[11, 12, 13],
#         [14, 15, 16]])

print(type(d))       # <class 'int'>
print(type(A + d))   # <class 'torch.Tensor'>
```

```
tensor([[11, 12, 13],
        [14, 15, 16]])
```

```
<class 'int'>
<class 'torch.Tensor'>
```

```
import torch

A = torch.tensor([[1, 2, 3],
                  [4, 5, 6]])   # shape (2,3)
b = torch.tensor([10, 20, 30])  # shape (3,)

print(A + b)
# tensor([[11, 22, 33],
#         [14, 25, 36]])
```

c02_pytorch04.py

## Numpy

```
import numpy as np

A = np.array([[1, 2, 3],
              [4, 5, 6]])   # shape (2,3)

d = 10
print(A + d)
# [[11 12 13]
#  [14 15 16]]

print(type(d))    # <class 'int'>
print(type(A+d))  # <class 'numpy.ndarray'>
```

```
[[11 22 33]
 [14 25 36]]
```

```
<class 'int'>
<class 'numpy.ndarray'>
```

```
import numpy as np

A = np.array([[1, 2, 3],
              [4, 5, 6]])   # shape (2,3)
b = np.array([10, 20, 30])  # shape (3,)

print(A + b)
# [[11 22 33]
#  [14 25 36]]
```

```
[[11 12 13]
 [14 15 16]]
```

# dim

## PyTorch

**IMPORTANT**

```
tensor(21.)
tensor(3.5000) tensor(1.) tensor(6.)
tensor(3.) tensor(1.8708)
```

```
import torch

B = torch.tensor([[1, 2, 3],
                  [4, 5, 6]], dtype=torch.float32)
# set dtype=float for convenient mean calculation

print(B.sum())                      # tensor(21.)
print(B.mean(), B.min(), B.max())
# tensor(3.5) tensor(1.) tensor(6.)

print(B.median(), B.std())          # tensor(3.5), tensor(1.8708)

# dim in PyTorch ------------------------------------------
print(B.sum(dim=0))   # column sums -> tensor([5., 7., 9.])
print(B.sum(dim=1))   # row sums    -> tensor([ 6., 15.])

# column-wise mean (dim=0)
col_means = B.mean(dim=0)
print("axis=0 (columns) mean:", col_means)
# tensor([2.5000, 3.5000, 4.5000])
# row-wise mean (dim=1)
row_means = B.mean(dim=1)
print("axis=1 (rows) mean:", row_means)   # tensor([2., 5.])
```

```
tensor([5., 7., 9.])
tensor([ 6., 15.])
```

```
axis=0 (columns) mean: tensor([2.5000, 3.5000, 4.5000])
```

```
axis=1 (rows) mean: tensor([2., 5.])
```

## Numpy

```
21
3.5 1 6
3.5 1.707825127659933
```

```
import numpy as np

B = np.array([[1, 2, 3],
              [4, 5, 6]])

print(B.sum())              # 21
print(B.mean(), B.min(), B.max())  # 3.5 1 6
print(np.median(B), np.std(B))     # 3.5, 1.7078

# axis ----------------------------------------
print(B.sum(axis=0))   # column sums -> [5 7 9]
print(B.sum(axis=1))   # row sums -> [ 6 15]

# column-wise mean (axis=0)
col_means = B.mean(axis=0)
print("axis=0 (columns) mean:", col_means)
# [2.5 3.5 4.5]

# row-wise mean (axis=1)
row_means = B.mean(axis=1)
print("axis=1 (rows) mean:", row_means)      # [2. 5.]
```

```
[5 7 9]
[ 6 15]
```

```
axis=0 (columns) mean: [2.5 3.5 4.5]
```

```
axis=1 (rows) mean: [2. 5.]
```

c02_pytorch05.py

# torch.std() vs np.std()

## PyTorch

```
import torch

A = torch.tensor([1, 2, 3, 4, 5, 6], dtype = float)

print(A.median(), A.std())          # tensor(3.), tensor(1.8708)
print(torch.median(A), torch.std(A))  # the same as the above
print(torch.std(A, unbiased=False))    # matches np.std
```

```
tensor(3., dtype=torch.float64) tensor(1.8708, dtype=torch.float64)
tensor(3., dtype=torch.float64) tensor(1.8708, dtype=torch.float64)
tensor(1.7078, dtype=torch.float64)
```

PyTorch computes the **sample standard deviation**:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu)^2}$$

$$s = \sqrt{\frac{1}{5}\,(2 * (2.5^2 + 1.5^2 + 0.5^2))}$$

## Numpy

```
import numpy as np

B = np.array([1, 2, 3, 4, 5, 6])

print(np.median(B), np.std(B))   # 3.5, 1.7078
```

```
3.5 1.707825127659933
```

NumPy computes the **population standard deviation**:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

$$\sigma = \sqrt{\frac{1}{6}\,(2 * (2.5^2 + 1.5^2 + 0.5^2))}$$

### Key Difference

The key difference lies in the denominator:

- NumPy: population($N$)
- PyTorch (default): sample ($N-1$)

# reshape( ), view( )

## PyTorch

```
import torch

C = torch.arange(12)     # tensor([0, 1, ..., 11])
print(C.reshape(3, 4))   # shape (3,4)
print(C.reshape(-1, 6)) # -1 lets PyTorch infer -> (2,6)
print(C.view(-1))        # flat view if possible
```

```
tensor([[ 0,  1,  2,  3],
        [ 4,  5,  6,  7],
        [ 8,  9, 10, 11]])
```

```
tensor([[ 0,  1,  2,  3,  4,  5],
        [ 6,  7,  8,  9, 10, 11]])
```

```
tensor([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11])
```

c02_pytorch06.py

## Numpy

```
import numpy as np

C = np.arange(12)        # [0..11]
print(C.reshape(3, 4))   # shape (3,4)
print(C.reshape(-1, 6)) # -1 lets NumPy infer -> (2,6)
print(C.ravel())         # flat view if possible
```

```
[[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]
```

```
[[ 0  1  2  3  4  5]
 [ 6  7  8  9 10 11]]
```

```
[ 0  1  2  3  4  5  6  7  8  9 10 11]
```

# reshape( ), view( )

PyTorch

```
# Create a contiguous tensor
A = torch.arange(12)
print("A:", A)

# Reshape works fine
print("A.reshape(3,4):\n", A.reshape(3,4))

# Make a non-contiguous tensor by transposing
B = A.reshape(3,4).t()   # transpose
C = A.reshape(3,4)
print("\nB (transposed):\n", B)
print("Is B contiguous?", B.is_contiguous())
print("Is C contiguous?", C.is_contiguous())

# view requires contiguous memory -> will raise an error
try:
    print(B.view(12))   # RuntimeError
except RuntimeError as e:
    print("B.view(12) failed:", e)

# reshape can handle non-contiguous tensors by copying
print("B.reshape(12):", B.reshape(12)) #  works
```

```
tensor([[ 0,  4,  8],
        [ 1,  5,  9],
        [ 2,  6, 10],
        [ 3,  7, 11]])
Is B contiguous? False
Is C contiguous? True
B.view(12) failed: view size is not compatible with input tensor's size and stride (at least one dimension spans across two contiguous subspaces). Use .reshape(...) instead.
B.reshape(12): tensor([ 0,  4,  8,  1,  5,  9,  2,  6, 10,  3,  7, 11])
```

- view → requires contiguous memory, otherwise it raises an error
- reshape → automatically creates a copy if needed and returns the result

c02_pytorch06.py

# reshape( ).t( ) vs flatten( )

PyTorch

```
E = torch.arange(12).reshape(3,4).t()  # non-contiguous
print("Is contiguous?", E.is_contiguous())

# flatten → always returns contiguous copy
print(E.flatten().is_contiguous())  # True

# view(-1) → only works if contiguous
try:
    print(E.view(-1))
except RuntimeError as e:
    print("view failed:", e)
```

```
Is contiguous? False
True
```

c02_pytorch06.py

```
view failed: view size is not compatible with input tensor's size and stride (at least one dimension spans a
cross two contiguous subspaces). Use .reshape(...) instead.
```

# view( )

```
import torch

t = torch.arange(12).view(3, 4)
print("reshaped (3x4):\n", t)
```

```
reshaped (3x4):
 tensor([[ 0,  1,  2,  3],
        [ 4,  5,  6,  7],
        [ 8,  9, 10, 11]])
```

# PyTorch

`B = A.reshape(3,4).t()` -> non-contiguous order in memory

**Original contiguous buffer**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**`reshape(3,4)` just change the view of memory:** –only the *view* (shape/stride) changes, memory stays the same.

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|

`B = A.reshape(3,4).t()` -> looks like a 4×3 matrix, but the elements are accessed in a non-contiguous order.

| 0 | 4 | 8 | 1 | 5 | 9 | 2 | 6 | 10 | 3 | 7 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|

## view

- Only if contiguous
- Fast (just metadata change)
- Raises error otherwise

## reshape

- If the data is stored in a simple, contiguous block (e.g., created with arange, zeros, etc.),

→ reshape just changes the shape.
→ This is a view (very fast, no new memory).

- If the data is not contiguous (e.g., after transpose, slicing with steps, etc.),

→ reshape cannot simply reinterpret memory.
→ It will make a copy internally to give you the desired shape.

# torch.tensor([1., 2.]) is a 1-D tensor (=rank 1) whose shape is (2,).

- In both NumPy and PyTorch, a 1-D vector has no explicit row or column orientation.
- Depending on the operation, PyTorch (and NumPy) implicitly treats it as a row or as a column.

```
import torch

A = torch.tensor([[1., 2.],
                  [3., 4.]])

b = torch.tensor([1., 2.])  # define vector b

row = b[None, :]           # shape: (1, 2) ← row vector
col = b[:, None]           # shape: (2, 1) ← column vector

# Equivalent forms
row2 = b.reshape(1, -1)    # (1, 2)
col2 = b.reshape(-1, 1)    # (2, 1)

# Matrix multiplication
print(b @ A)               # (2,) @ (2,2) → (2,)   (behaves like a row in this context)
print(A @ b)               # (2,2) @ (2,) → (2,)   (behaves like a column in this context)

print(row @ A)             # (1,2) @ (2,2) → (1,2)
print(A @ col)             # (2,2) @ (2,1) → (2,1)
```

```
tensor([ 7., 10.])
tensor([ 5., 11.])
tensor([[ 7., 10.]])
tensor([[ 5.],
        [11.]])
```

```
row = b[None, :]
```

None tells PyTorch to 'add a new axis (dimension).'

c02_pytorch07.py

# torch.stack( )

torch.stack( ) → adds a new axis.

```
import torch

X = torch.tensor([1, 2, 3])
Y = torch.tensor([4, 5, 6])

print(torch.stack([X, Y], dim=0))  # [[1, 2, 3]
                                   #  [4, 5, 6]]

print(torch.stack([X, Y], dim=1))  # [[1, 4],
                                   #  [2, 5],
                                   #  [3, 6]]

print(torch.vstack([X, Y]))        # [[1, 2, 3],
                                   #  [4, 5, 6]]

print(torch.hstack([X, Y]))        # [1, 2, 3, 4, 5, 6]
```

```
tensor([[1, 2, 3],
        [4, 5, 6]])

tensor([[1, 4],
        [2, 5],
        [3, 6]])

tensor([[1, 2, 3],
        [4, 5, 6]])

tensor([1, 2, 3, 4, 5, 6])
```

```
import torch

X = torch.tensor([1, 2, 3])
Y = torch.tensor([4, 5, 6])
print(X.shape, Y.shape)

a = torch.stack([X, Y], dim=0)
b = torch.stack([X, Y], dim=1)

print(a.shape)  # torch.Size([2, 3])
print(b.shape)  # torch.Size([3, 2])
```

```
torch.Size([3]) torch.Size([3])
torch.Size([2, 3])
torch.Size([3, 2])
```

c02_pytorch08.py

# Linear algebra

c02_pytorch09.py

```
import torch

A = torch.tensor([[2., 1.],
                  [1., 3.]])
b = torch.tensor([1., 2.])

# Solve A x = b
x = torch.linalg.solve(A, b)
print("solution:", x)

# Matrix multiply (dot product)
M = torch.tensor([[1, 2],
                  [3, 4]], dtype=torch.float64)
N = torch.tensor([[10, 20],
                  [30, 40]], dtype=torch.float64)

print(M * N)          # element-wise multiplication
print(M @ N)          # matrix multiplication
print(torch.matmul(M, N))  # matrix multiplication (same as @)

# Eigenvalues & eigenvectors
eigvals, eigvecs = torch.linalg.eig(M)
print("eigenvalues:", eigvals)
print("eigenvectors:\n", eigvecs)
```

```
solution: tensor([0.2000, 0.6000])
```

```
tensor([[ 10.,  40.],
        [ 90., 160.]], dtype=torch.float64)
```

```
tensor([[ 70., 100.],
        [150., 220.]], dtype=torch.float64)
```

```
tensor([[ 70., 100.],
        [150., 220.]], dtype=torch.float64)
```

```
eigenvalues: tensor([-0.3723+0.j,  5.3723+0.j], dtype=torch.complex128)
eigenvectors:
 tensor([[-0.8246+0.j, -0.4160+0.j],
         [ 0.5658+0.j, -0.9094+0.j]], dtype=torch.complex128)
```

# Converting between Numpy and PyTorch Tensor

```python
import torch
import numpy as np

np_array = np.array([1, 2, 3])

# Numpy -> Tensor
tensor_from_np = torch.from_numpy(np_array)
print("Tensor from numpy:", tensor_from_np)

# Tensor -> Numpy
back_to_np = tensor_from_np.numpy()
print("Back to numpy:", back_to_np)
```

```
Tensor from numpy: tensor([1, 2, 3])
Back to numpy: [1 2 3]
```

c02_pytorch10.py

# Using GPU

```
# device (use GPU if available)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

a = torch.tensor([1., 2.], device=device)
b = torch.tensor([1., 2.])
print(a)
print(b)
```

```
tensor([1., 2.], device='cuda:0')
tensor([1., 2.])
```

c02_pytorch10.py

CPU


GPU

# torch.linspace(0, 1, N)

```
import torch

N =5
x = torch.linspace(0, 1, N)
print(x)

y = torch.linspace(0, 1, N).unsqueeze(0)
print(y)

z = torch.linspace(0, 1, N).unsqueeze(1)
print(z)

print(x.shape, y.shape, z.shape)
```

```
tensor([0.0000, 0.2500, 0.5000, 0.7500, 1.0000])

tensor([[0.0000, 0.2500, 0.5000, 0.7500, 1.0000]])

tensor([[0.0000],
        [0.2500],
        [0.5000],
        [0.7500],
        [1.0000]])

torch.Size([5]) torch.Size([1, 5]) torch.Size([5, 1])
```

# Optimizers in torch.optim

**Optim.SGD**

- The most **basic** Stochastic Gradient Descent.
- Supports options like momentum and nesterov

**Optim.RMSProp**

- Often used for training RNNs

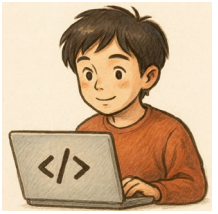
**Optim.Adam**

- One of the most **widely used** optimizers.
- Based on Adaptive Moment Estimation.
- Combines the benefits of momentum and RMSProp.

**Optim.AdamW**

- A variant of Adam with a proper implementation of weight decay.
- Standard choice for Transformer models such as BERT.

# Simplest Example (1): Linear Regression (1D Loss)



We aim to learn the parameters $a, b \in \mathbb{R}$ of a linear function:

$$y = f(x) = ax + b$$

## Training Data

Given input samples $\{x_i\}_{i=1}^{N}$, we generate targets:

$$y_i = a^* x_i + b^* + \epsilon_i$$

where $a^*, b^*$ are the true parameters and $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ is Gaussian noise.

## Model Parameters

We initialize parameters randomly:

$$a \sim \mathcal{N}(0, 1), \quad b \sim \mathcal{N}(0, 1).$$

## Prediction

For each input $x_i$, the prediction is:

$$\hat{y}_i = ax_i + b.$$

# Simplest Example (1): Linear Regression (1D Loss)

## Loss Function

We minimize the mean squared error (MSE):

$$\mathcal{L}(a, b) = \frac{1}{N} \sum_{i=1}^{N} \left(\hat{y}_i - y_i\right)^2.$$

## Optimization

Parameters are updated with gradient descent (SGD):

$$a \leftarrow a - \eta \frac{\partial \mathcal{L}}{\partial a}, \quad b \leftarrow b - \eta \frac{\partial \mathcal{L}}{\partial b},$$

where $\eta > 0$ is the learning rate.

## Result

After sufficient iterations, the learned parameters

$$(a, b) \approx (a^*, b^*)$$

converge to the true coefficients of the underlying linear function.

# Simplest Example (1A): Linear Regression (1D Loss)

c02_pytorchEx01A.py

```
import torch

torch.manual_seed(0)   # Reproducibility

# 1) Ground-truth (random) + data
a_star = torch.randn(1)          # true slope
b_star = torch.randn(1)          # true y-intercept
N = 200
x = torch.linspace(-5, 5, N).unsqueeze(1) # .unsqueeze(1) adds a new dimension at index 1 : making a column vector
noise_std = 0.5
y = a_star * x + b_star + noise_std * torch.randn(N, 1)

# 2) Parameters to learn (scalars with grad)
a = torch.randn(1, requires_grad=True)
b = torch.randn(1, requires_grad=True)

# 3) Optimizer (Adam tends to be steadier than plain SGD here)
opt = torch.optim.Adam([a, b], lr=0.05)

# 4) Training loop
for _ in range(3000):            # more steps → steadier convergence
    y_hat = a * x + b
    loss = torch.mean((y_hat - y) ** 2)

    opt.zero_grad()              # clear old gradients
    loss.backward()              # compute ∂L/∂a, ∂L/∂b
    opt.step()                   # update a, b

print(f"True:    a* = {a_star.item():.4f}, b* = {b_star.item():.4f}")
print(f"Learned: a  = {a.item():.4f}, b  = {b.item():.4f}")
print(f"MSE: {loss.item():.4f}")
```

```
True:    a* = 1.5410, b* = -0.2934
Learned: a  = 1.5372, b  = -0.2266
MSE: 0.2349
```

# Simplest Example (1B): Linear Regression (1D Loss)

## With nn.Module

c02_pytorchEx01B.py

```
import torch
import torch.nn as nn

torch.manual_seed(0)   # Reproducibility

# 1) Ground-truth (random) + data
a_star, b_star = torch.randn(1), torch.randn(1)
N = 100
x = torch.linspace(-5, 5, N).unsqueeze(1)  # .unsqueeze(1) adds a new dimension at index 1 : making a column vector
y = a_star * x + b_star + 0.5 * torch.randn(N, 1)

# 2) Model, loss, optimizer
model = nn.Linear(1, 1)              # y ≈ ax + b   (nn.Linear is a subclass of nn.Module), model is an object.
criterion = nn.MSELoss()             # MSELoss is a class and criterion is the instance of MSELoss class
optimizer = torch.optim.SGD(model.parameters(), lr=0.05)  #torch.optim.SGD is a class. Optimizer is an object.

# 3) Train
for _ in range(3000):                 # more steps → steadier convergence
    y_hat = model(x)
    loss = criterion(y_hat, y)
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

a, b = model.weight.item(), model.bias.item()
print(f"True: a* = {a_star.item():.4f}, b* = {b_star.item():.4f}")
print(f"Learned: a = {a:.4f}, b = {b:.4f}")
print(f"MSE: {loss.item():.4f}")
```

Let `criterion` be an instance of `nn.MSELoss`:

$$\texttt{criterion} = \texttt{nn.MSELoss()}.$$

## Callable Object via `__call__`

In Python, when we write

$$\texttt{loss} = \texttt{criterion}(y_{\text{hat}}, y),$$

this does not call the constructor again. Instead, it invokes the special method

$$\texttt{criterion.\_\_call\_\_(y\_hat, y)}.$$

## Delegation to `forward`

For PyTorch modules, the method `__call__` internally delegates to the `forward` function:

$$\texttt{criterion.\_\_call\_\_(y\_hat, y)} \longrightarrow \texttt{criterion.forward}(y\_\text{hat}, y).$$

## Mean Squared Error Computation

Specifically, for `nn.MSELoss`, the forward method computes the mean squared error:

$$\mathcal{L}(y_{\text{hat}}, y) = \frac{1}{N} \sum_{i=1}^{N} (y_{\text{hat},i} - y_i)^2 .$$

## Summary

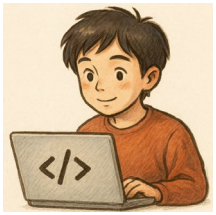
Thus,

$$\texttt{loss} = \texttt{criterion}(y_{\text{hat}}, y)$$

is equivalent to evaluating the MSE loss between the prediction $y_{\text{hat}}$ and the target $y$.

IMPORTANT

# Simple Example (2): Linear Regression



Let the input data matrix, weight vector, and target value be

$$X \in \mathbb{R}^{N\times d}, \quad w \in \mathbb{R}^{d\times 1}, \quad t \in \mathbb{R}^{N\times 1},$$

respectively. Here, N = 5 and d = 3 (5 samples, 3 features).

Predictions without bias:

$$\hat{y} = Xw.$$

## Loss Function (MSE: Mean Squared Error)

$$L(w) = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - t_i)^2 = \frac{1}{N}\|Xw - t\|_2^2.$$

If we use a scalar target $c$ (broadcasted to all samples), then $t_i = c$ and:

$$L(w) = \frac{1}{N}\sum_{i=1}^{N}(x_i^\top w - c)^2.$$

target

$$t \in \mathbb{R}^{N\times 1}, \quad t = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

## Gradients

$$\nabla_w L(w) = \frac{2}{N}X^\top(Xw - t),$$

# Simple Example (2): Linear Regression

```python
import torch

# device
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# inputs
x = torch.randn(5, 3, device=device)   # 5 samples, each with 3 features
target = torch.tensor([[0.5]]*5, device=device)  # The shape of target vector (all 0.5) is (5,1).

# parameter to learn
w = torch.randn(3, 1, device=device, requires_grad=True) # random values are sampled from N(0,1)
                # requires_grad=True tells autograd to track this tensor and compute gradients

# optimizer
optimizer = torch.optim.SGD([w], lr=0.1)

for step in range(20):
    pred = x @ w                          # The shape of pred is (5,1). @ does matrix multiplication.
    loss = ((pred - target)**2).mean()  # MSE loss

    # autograd (= automatically computes gradients of tensors)
    optimizer.zero_grad()   # Reset gradients from the previous step (set .grad to zero)
    loss.backward()         # Compute gradients of loss (=d(loss)/dw ) w.r.t parameters → stored in w.grad
    optimizer.step()        # Update w using gradients: w = w - lr * w.grad

    if step % 5 == 0:  # print every 5 steps
        print(f"Step {step:2d} | loss {loss.item():.4f} | w = {w.detach().T}")
        # .detach() breaks the tensor away from the autograd graph (so it won't track gradients).

print("Final w:", w.detach().T)
```

```
Step  0 | loss 0.8743 | w = tensor([[ 0.6639, -0.4362,  0.3074]], device='cuda:0')
Step  5 | loss 0.1398 | w = tensor([[ 0.2987, -0.5716,  0.0984]], device='cuda:0')
Step 10 | loss 0.0936 | w = tensor([[ 0.2158, -0.5652,  0.0095]], device='cuda:0')
Step 15 | loss 0.0859 | w = tensor([[ 0.1982, -0.5498, -0.0406]], device='cuda:0')
Final w: tensor([[ 0.1977, -0.5432, -0.0687]], device='cuda:0')
```

c02_pytorchEx02A.py

# Simple Example (2): Linear Regression using torch.nn

c02_pytorchEx02B.py

```python
import torch
import torch.nn as nn

# device (use GPU if available)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# inputs
x = torch.randn(5, 3, device=device)                # 5 samples, 3 features each (shape: 5x3)
target = torch.full((5, 1), 0.5, device=device)     # target values: constant 0.5 (shape: 5x1)

# model: y = x @ w, no bias (same as the original manual implementation)
model = nn.Linear(3, 1, bias=False).to(device)

# optimizer & loss function
optimizer = torch.optim.SGD(model.parameters(), lr=0.1)
loss_fn = nn.MSELoss()

for step in range(20):
    optimizer.zero_grad(set_to_none=True)   # reset gradients
    pred = model(x)                         # forward pass (output shape: 5x1)
    loss = loss_fn(pred, target)            # mean squared error loss
    loss.backward()                         # backpropagation
    optimizer.step()                        # parameter update

    if step % 5 == 0:
        with torch.no_grad():
            # model.weight has shape (1,3), so no need to transpose for printing
            print(f"Step {step:2d} | loss {loss.item():.4f} | w = {model.weight.detach()}")

with torch.no_grad():
    print("Final w:", model.weight.detach())
    print(target.shape)
```

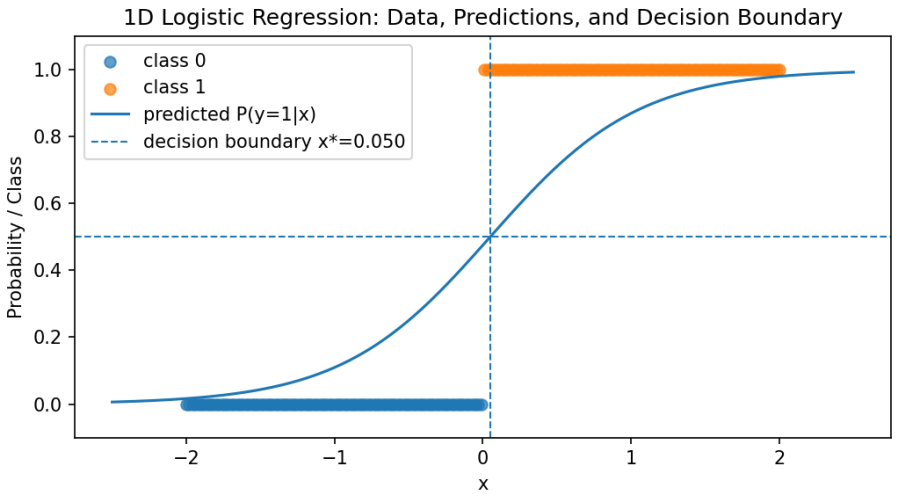
`with torch.no_grad():`→ temporarily disables gradient tracking when you're only reading values (like model outputs or weights).

`Model.weight` → a learnable parameter in PyTorch. That means it is a torch.Tensor with `requires_grad=True` so that gradients can be computed during backpropagation.

```
Step  0 | loss 0.4216 | w = tensor([[-0.4224,  0.5199, -0.5011]], device='cuda:0')
Step  5 | loss 0.2786 | w = tensor([[-0.2865,  0.5620, -0.3276]], device='cuda:0')
Step 10 | loss 0.2443 | w = tensor([[-0.2172,  0.5417, -0.2330]], device='cuda:0')
Step 15 | loss 0.2291 | w = tensor([[-0.1780,  0.5005, -0.1719]], device='cuda:0')
Final w: tensor([[-0.1569,  0.4636, -0.1352]], device='cuda:0')
torch.Size([5, 1])
```

We use `.detach()` to safely look at the parameter values without gradient tracking.

# Simple Example (3): Binary Classification




## 1. Dataset (1D line)

We construct $N$ (in our case, $N = 200$) evenly spaced points on a 1D line and assign binary labels via a step rule:

$$x_i \in \mathbb{R}, \qquad x_i \text{ linspace in } [-2, 2], \qquad y_i = \mathbf{1}\{\, x_i > 0 \,\} \in \{0, 1\}, \quad i = 1, \ldots, N.$$

In code: `x = torch.linspace(-2,2,N).unsqueeze(1)`, `y = (x[:,0] > 0).float()`.

## 2. Model (Logistic Regression)

A single linear (affine) function maps input $x \in \mathbb{R}$ to a scalar:

$$z_i = wx_i + b, \qquad \boxed{w, b \in \mathbb{R}.}$$

The sigmoid (logistic) function gives the class probability

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \qquad \hat{p}_i = \Pr(y_i = 1 \mid x_i) = \sigma(z_i) = \sigma(wx_i + b).$$

# Simple Example (3): Binary Classification

## 3. Loss: Binary Cross-Entropy (with logits)

For a label $y_i \in \{0, 1\}$ and logit $z_i$,

$$\ell(z_i, y_i) \;=\; -\Big(y_i \log \sigma(z_i) + (1 - y_i) \log\big(1 - \sigma(z_i)\big)\Big).$$

PyTorch's `BCEWithLogitsLoss` implements a numerically stable form. The empirical risk minimized during training is the mean loss:

$$\mathcal{L}(w, b) \;=\; \frac{1}{N} \sum_{i=1}^{N} \ell\big(wx_i + b,\; y_i\big).$$

## 4. Gradients (1D closed form)

Using $\frac{d}{dz}\ell(z, y) = \sigma(z) - y$, we get

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{1}{N} \sum_{i=1}^{N} \big(\sigma(wx_i + b) - y_i\big)\, x_i, \qquad \frac{\partial \mathcal{L}}{\partial b} = \frac{1}{N} \sum_{i=1}^{N} \big(\sigma(wx_i + b) - y_i\big).$$

# Why BCE Loss and PyTorch's Stable Form Are Equivalent

## 1. Problem: Numerical Instability

If $z$ is a very large positive number, then $e^{-z} \approx 0$, so computing $\log(1 + e^{-z})$ may cause underflow. If $z$ is a very large negative number, then $e^{z}$ becomes extremely small, and computing $\log(1 + e^{z})$ may cause overflow/underflow issues.

In other words, using the expression $e^{-z}$ directly for computing sigmoid function, $\sigma(z) = \frac{1}{1+e^{-z}}$ can easily lead to floating-point precision problems, producing NaN or Inf values.

## 2. Standard BCE Loss (with logits)

For binary classification with $y \in \{0, 1\}$ and a logit $z \in \mathbb{R}$, the Binary Cross-Entropy (BCE) loss is

$$\ell_{\mathrm{BCE}}(z, y) = -\Big(y \log \sigma(z) + (1 - y) \log \big(1 - \sigma(z)\big)\Big), \tag{1}$$

where the sigmoid is $\sigma(z) = \frac{1}{1+e^{-z}}$.

Expanding the terms:

$$-\log \sigma(z) = \log(1 + e^{-z}), \qquad -\log(1 - \sigma(z)) = \log(1 + e^{z}), \tag{2}$$

so

$$\ell_{\mathrm{BCE}}(z, y) = y \log(1 + e^{-z}) + (1 - y) \log(1 + e^{z}). \tag{3}$$

## 3. Using the Softplus Function

Define the softplus:

$$\text{softplus}(z) := \log(1 + e^z). \tag{4}$$

Then (3) can be rewritten as

$$\ell_{\text{BCE}}(z, y) = y \ \text{softplus}(-z) + (1 - y) \ \text{softplus}(z). \tag{5}$$

## 4. Key Identity

Note that

$$\text{softplus}(-z) = \log(1 + e^{-z}) = \log(1 + e^z) - z = \text{softplus}(z) - z. \tag{6}$$

Substituting (6) into (5) gives

$$\ell_{\text{BCE}}(z, y) = \text{softplus}(z) - z\, y. \tag{7}$$

## 5. Stable Decomposition of Softplus

A numerically stable identity for the softplus is

$$\text{softplus}(z) = \log(1 + e^z) = \max(z, 0) + \log\left(1 + e^{-|z|}\right). \tag{8}$$

Proof (case analysis):

- If $z \geq 0$: $\max(z, 0) = z$, $|z| = z$, so

$$z + \log(1 + e^{-z}) = \log(e^z(1 + e^{-z})) = \log(1 + e^z).$$

- If $z < 0$: $\max(z, 0) = 0$, $|z| = -z$, so

$$0 + \log(1 + e^z) = \log(1 + e^z).$$

Thus, (8) holds for all $z$.

## 6. PyTorch's Stable BCE Loss

Combining (7) and (8),

$$\ell_{\mathrm{BCE}}(z, y) = \left[\max(z, 0) + \log(1 + e^{-|z|})\right] - z\,y = \max(z, 0) - z\,y + \log(1 + e^{-|z|}). \tag{9}$$

## 7. Final Summary

The following three forms are mathematically identical:

$$\ell(z, y) = -\left[y \log \sigma(z) + (1-y) \log(1-\sigma(z))\right] = \mathrm{softplus}(z) - zy = \max(z, 0) - zy + \log(1+e^{-|z|}). \tag{10}$$

The last form (9) is used in `BCEWithLogitsLoss` because it is numerically stable for large $|z|$ (avoiding overflow/underflow).

# Simple Example (3): Binary Classification

```
import torch

# Toy dataset (1D line) Label = 1 if x > 0, else 0
N = 200
x = torch.linspace(-2, 2, N).unsqueeze(1)           # shape: (N,1)
y = (x[:,0] > 0).float()                            # step function

# Logistic regression model
model = torch.nn.Linear(1, 1)                       # w*x + b
criterion = torch.nn.BCEWithLogitsLoss()
optimizer = torch.optim.SGD(model.parameters(), lr=0.1)

# Training
for step in range(100):
    logits = model(x).squeeze(1)                    # raw scores
    loss = criterion(logits, y)                     # binary CE loss

    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

    if step % 20 == 0:
        with torch.no_grad():
            preds = (logits > 0).float()            # threshold at 0
            acc = (preds == y).float().mean().item()
        print(f"step={step:3d}  loss={loss.item():.4f}  acc={acc:.3f}")
```

```
step=  0  loss=0.8361  acc=0.500
step= 20  loss=0.4480  acc=0.805
step= 40  loss=0.3181  acc=0.920
step= 60  loss=0.2595  acc=0.955
step= 80  loss=0.2259  acc=0.970
```

c02_pytorchEx03.py

# Thank you!